SOFTWARE ENGINEERING LABORATORY

LAB–5

DATA FLOW DIAGRAM (DFD)

LIBRARY MANAGEMENT SYSTEM

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Structured System Analysis and Design

1. Aim

To prepare Data Flow Diagrams for the Library Management System to show how data enters the system, how it is processed, where it is stored, and how information is returned to users.

2. Introduction

A Data Flow Diagram (DFD) is a graphical representation of the movement of data through an information system. It shows external entities, processes, data stores, and data flows. A DFD focuses on the flow and transformation of information rather than the sequence of program execution.

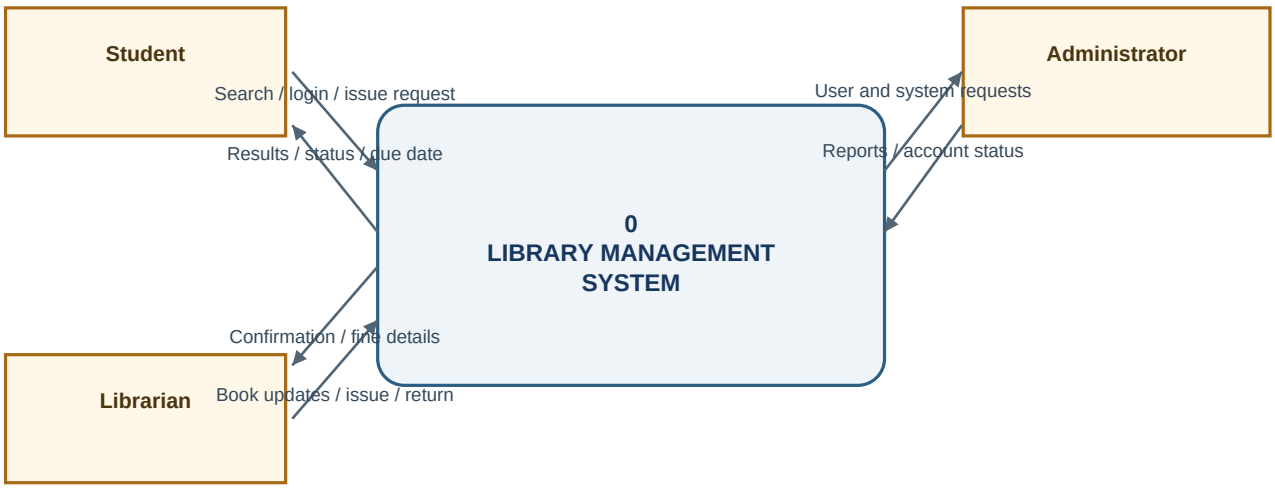
3. DFD Symbols

Symbol / Element	Meaning	Representation in this document
External Entity	A person or external system that sends data to or receives data from the system.	Rectangle – Student, Librarian, Administrator
Process	Transforms input data into output data.	Rounded rectangle – numbered process
Data Store	Stores data for later use.	Open-ended / double-line style store
Data Flow	Shows data moving between entities, processes, and stores.	Directed arrow with a label

4. Level 0 DFD – Context Diagram

The Level 0 DFD represents the entire Library Management System as one process. It shows the three external entities—Student, Librarian, and Administrator—and the information exchanged with the system.

LEVEL 0 – CONTEXT DIAGRAM

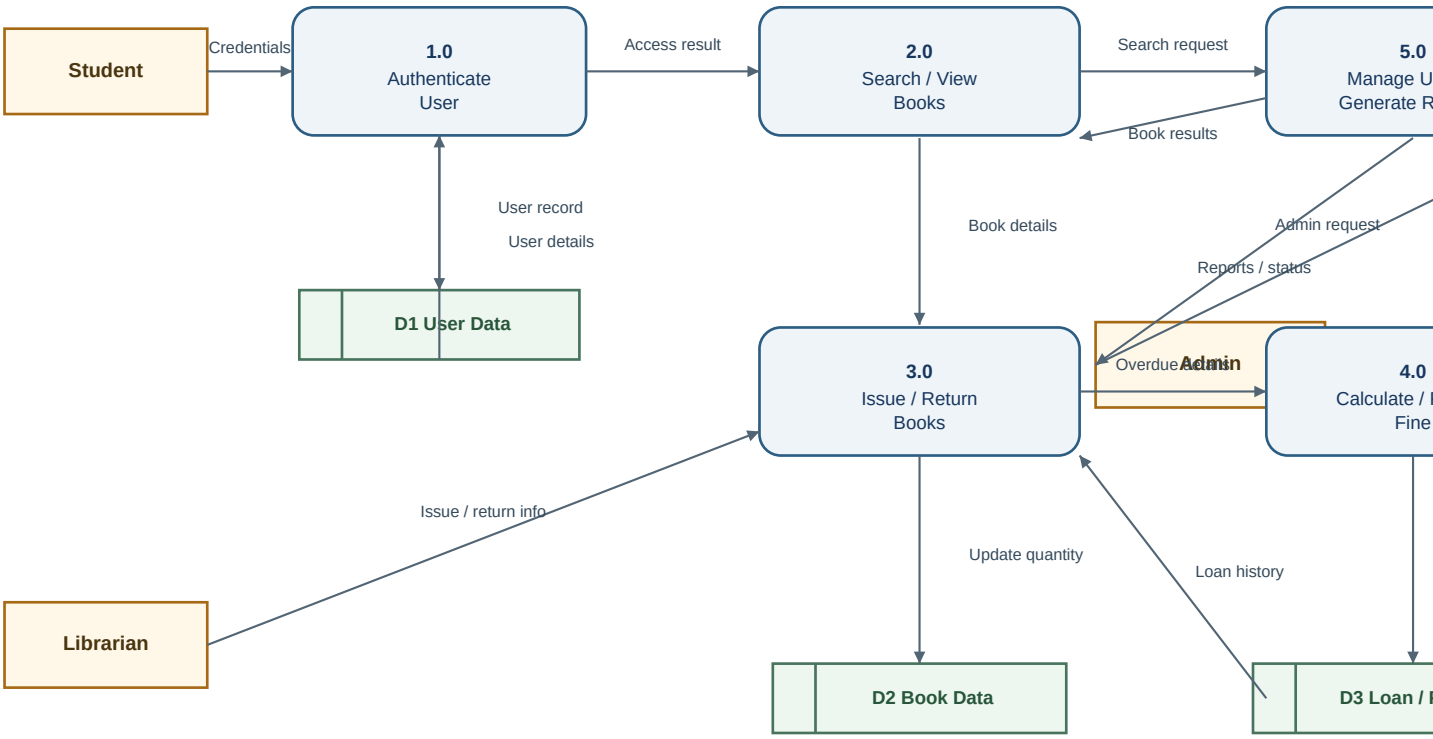


External entities interact with one central system process.

5. Level 1 DFD – System Decomposition

The Level 1 DFD breaks the main system into five major processes: user authentication, book search, book issue/return, fine management, and user/report management. It also shows the main data stores used by these processes.

LEVEL 1 – DECOMPOSITION OF LIBRARY MANAGEMENT SYSTEM



D1: User Data D2: Book Data D3: Loan and Fine Data

6. Description of Processes and Data Stores

ID	Process	Function
1.0	Authenticate User	Receives login credentials, validates them against User Data, and returns an authentication result.
2.0	Search / View Books	Accepts search criteria, reads book details and availability from Book Data, and returns matching results.
3.0	Issue / Return Books	Checks book and member details, creates or updates loan records, and updates book availability.
4.0	Calculate / Record Fine	Checks overdue information, calculates applicable fines, and stores fine and payment status.
5.0	Manage Users / Generate Reports	Allows the administrator to manage user accounts and request reports using stored system data.

Data Store	Stored Information
D1 – User Data	Student, librarian, and administrator account details, roles, and authentication-related information.
D2 – Book Data	Book ID, ISBN, title, author, category, total quantity, and available quantity.
D3 – Loan / Fine Data	Issue records, due dates, return dates, overdue details, fine amounts, and payment status.

7. Conclusion

The Level 0 and Level 1 DFDs describe the flow of information in the Library Management System. The context diagram identifies external entities and their interaction with the system, while the Level 1 diagram explains the major internal processes and data stores. These diagrams help in understanding system boundaries, data movement, and database requirements.

END OF LAB–5